

Wi-Fi Performance

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1 INTRODUCTION

In this laboratory, we will assess the performance of Wi-Fi connections under different conditions. The primary objective is to determine the "Theoretical Maximum Throughput" (of the Application Layer) or " TMT_{APP} " which, from this point on, will be called **Goodput**.

The study will focus on two distinct network setups, each tested under multiple scenarios. For each scenario, we collect and analyze key statistical metrics, including the minimum, maximum, average, and standard deviation of network performance indicators, such as throughput latency and packet loss.

To ensure a comprehensive evaluation, we will begin by hypothesizing the expected outcomes based on theoretical considerations (Section 2). We will then provide a detailed description of the analysed setups, conduct the experiments, systematically collect data, and generate graphs to enhance visual representation (Section 3). Finally, we will compare the measured results with our initial predictions, interpret and discuss the findings, and draw conclusions about Wi-Fi performance under varying conditions (Section 4).

Beyond evaluating network performance, this study aims to highlight the impact of environmental factors, interference, and device configurations on Wi-Fi efficiency. By systematically varying conditions such as signal strength, distance, and concurrent network activity, we seek to identify key factors that influence Goodput and overall network stability. In particular, we will conduct multiple tests by varying both the packet size and transmission rate to assess their effects on throughput, latency, and packet loss. The insights gained from this analysis may contribute to a better understanding of wireless network optimization strategies, aiding in both practical deployments and further research in the field.

All the data used for the calculations in this document can be found at the following link:

- [Google Drive Folder](#)

Clicking on the link will redirect you to the folder containing the necessary datasets and resources.

2 TOOLS AND THEORY

2.1 Tools used

Our analysis relies on two essential tools:

- **iperf3**, an open-source utility designed for measuring network throughput and performance
- **Wireshark**, a widely used open-source packet analyser for inspecting and analysing network traffic

2.2 Goodput formula

To evaluate network performance, we use the formula:

$$G = \mu \cdot C \quad (1)$$

where:

- G is the expected goodput.
- μ is a proportional efficiency factor ($0 < \mu < 1$), defined as:

$$\mu = \frac{\beta}{\alpha + \beta} \quad (2)$$

where β represents the application datagram size, and α is the total protocol overhead above the MAC layer. This factor is highly dependent on packet size and its value moves from closer to 0 towards 1 as β increases. The standard datagram size is 1500B

- C represents the capacity of the network channel, serving as the upper limit fixed a certain network structure

Among these parameters, C is the most easily obtainable.

By executing the following commands on both stations:

```
ethtool eth0  
iwconfig wlan0
```

we obtain the same results across both devices:

- $C_{eth} = 1000 \text{ Mb/s}$
- $C_{wifi} = 780 \text{ Mb/s}$

The efficiency factor μ differs for Ethernet and Wi-Fi.

For standard Ethernet communication both TCP and UDP exhibit high efficiency, typically around 0.95, increasing to 0.98–0.99 for larger packet sizes. For a standard 1500B packet, we take:

- $\mu_{eth, TCP} = 0.949$
- $\mu_{eth, UDP} = 0.957$

Wi-Fi efficiency varies significantly based on the protocol. Older standards, such as 802.11g and below, typically achieve around 50% efficiency (more precisely, $\mu < 0.50$ for TCP and $\mu < 0.55$ for UDP), whereas modern protocols provide substantially higher values.

For 802.11ac, efficiency ranges between 70–80%, with UDP performing slightly better than TCP. Efficiency also increases with packet size. For a standard 1500B packet in 802.11ac, we consider:

- $\mu_{wifi, TCP} = 0.70$
- $\mu_{wifi, UDP} = 0.75$

2.3 Theoretical results applied to setups

In a **fully Wi-Fi setup**, any measured result until this point **must be divided by two**. This is because each sender-receiver communication actually involves two separate Wi-Fi connections: one from the sender to the access point and another from the access point to the receiver:

- $G = \frac{\mu_{wifi, TCP} \cdot C_{wifi}}{2}$
- $G = \frac{\mu_{wifi, UDP} \cdot C_{wifi}}{2}$

A completely different story in an **hybrid setup**, where the **Goodput is inherently constrained by the bottleneck**, which in this case is always the **Wi-Fi connection**

- $G = \mu_{wifi,TCP} \cdot C_{wifi}$
- $G = \mu_{wifi,UDP} \cdot C_{wifi}$

In the end, Expected Goodput should look like this:

Setup	Protocol	Expected Goodput
Both Wi-Fi	TCP	$G = 273Mb/s$
	UDP	$G = 292.5Mb/s$
Mixed Mode	TCP	$G = 546Mb/s$
	UDP	$G = 585Mb/s$

Table 1: Expected Goodput computed with values of our previous considerations

3 SETUP AND SCENARIOS

All tests were conducted in a controlled environment, ensuring that the test stations were the only devices connected to the router.

The experimental setup consisted of three primary components:

- (1) **Client station:** A first PC, responsible for sending data, running Kali Linux OS virtualised on Windows.
- (2) **Server station:** A second PC, responsible for receiving data, also running Kali Linux OS virtualised on Windows.
- (3) **Access point:** A Fritz!Box 7590 router, acting as the intermediary between the client and server, with a nominal full-duplex Ethernet speed of 1 Gb/s and supporting Wi-Fi protocols 802.11b/g/n/ac, with a maximum nominal speed of 1.7 Gb/s and a bandwidth of 2.4/5 GHz.

3.1 Setups

Two primary setups were examined:

- **Both Wi-Fi setup:** Both the client and server were connected to the access point via Wi-Fi, using the 802.11ac protocol.

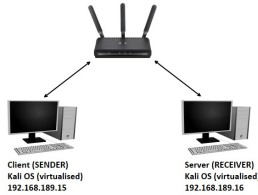


Figure 1: Scheme of setup "Both Wi-Fi"

- **Mixed mode setup:** The server remained connected via Wi-Fi (802.11ac), while the client was connected via Ethernet in full-duplex mode, using a CAT5 cable that ran at 1000BASE-T (Gigabit connection).

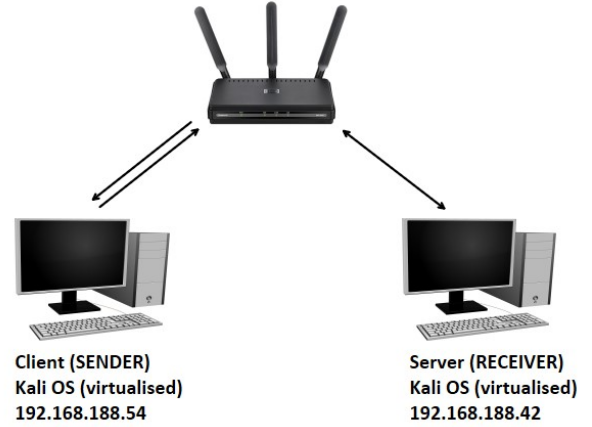


Figure 2: Scheme of setup "Mixed Mode"

3.2 Scenarios and variable parameters

For each setup, three key parameters were varied to assess their impact on Goodput measurements:

- **Transport layer protocol:** Both TCP and UDP were tested.
- **Bandwidth allocation:** Data transmission was evaluated at bitrates of 1 Mb/s, 100 Mb/s, 400 Mb/s, and 600 Mb/s.
- **Packet size at the application layer:** Tests were conducted with packet sizes of 100 B, 1500 B, 5000 B, and 65,000 B.

In total, 32 scenarios were tested for each setup (2 protocols $\times$ 4 bandwidth levels $\times$ 4 packet sizes). Each scenario was repeated 10 times to ensure statistically significant results. The collected data included average, minimum, maximum, and standard deviation values for receiver throughput.

3.3 Key Assumption

In this study, any latency introduced by virtualisation was considered negligible and did not significantly impact the throughput measurements.

4 RESULTS

The experiment conducted aimed to analyze the performance of the networks in terms of expected theoretical goodput and real goodput. The tests have been performed with UDP and TCP buffers in a wide range of values both for transmitted bandwidth and buffer size. The results will be presented underlining differences between setups, buffer size, and bandwidth.

4.1 Setups comparison with TCP

In setup "mixed mode", considering a mixed network that involves both wifi connections and LAN connections, the goodput is sensibly higher due to the higher stability of LAN connections. The wifi connection is in any case the bottleneck of the network in setup "both wifi mode" the real goodput is lower since both connections are prone to collisions and are affected by the lower reliability of wifi links (Figure 3)

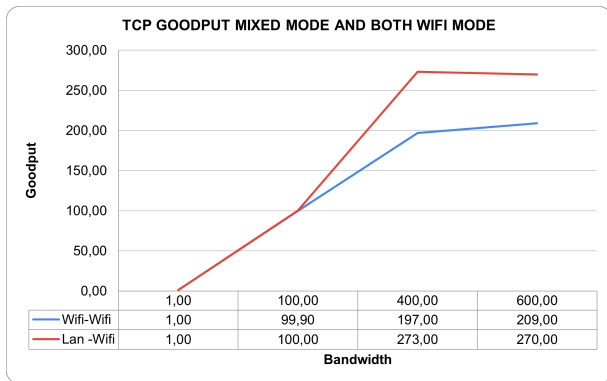


Figure 3: Comparison between goodput using TCP with "Both Wi-Fi Mode" and "Mixed Mode"

4.2 Buffer size and bandwidth comparison

The analysis reveals that the most notable differences between the two network configurations arise from the configured bandwidth and buffer size. These factors play a crucial role in determining the overall performance and efficiency of data transmission. The protocol used for the test is UDP and the overall results showed in any case higher goodput and lower packet loss using the mixed mode. The mixed mode showed better overall results and goodput increases until a threshold is reached only if a 65000B buffer is used to perform the test. Smaller buffers result in smaller goodput.

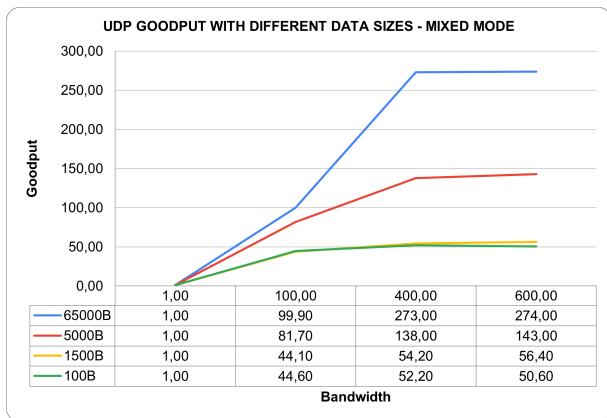


Figure 4: UDP goodput with different application buffer sizes and different bandwidth in "Mixed Mode"

Packet loss in mixed mode is negligible if a 65000B buffer is used, on the contrary the loss sensibly increases until 80% using buffers of 100B and 1500B

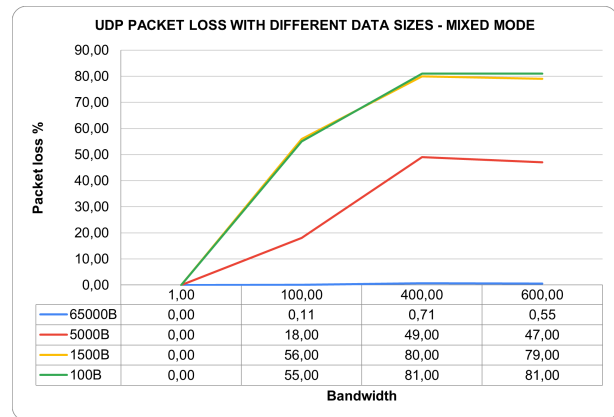


Figure 5: UDP packet loss with different application buffer sizes and different bandwidth in "Mixed Mode"

The "both wifi" setup shows the limitations of the wifi links in terms of link quality, demonstrating worse overall performances. Also using a 65000B buffer the goodput decreases when reaching a bandwidth of 600Mbps

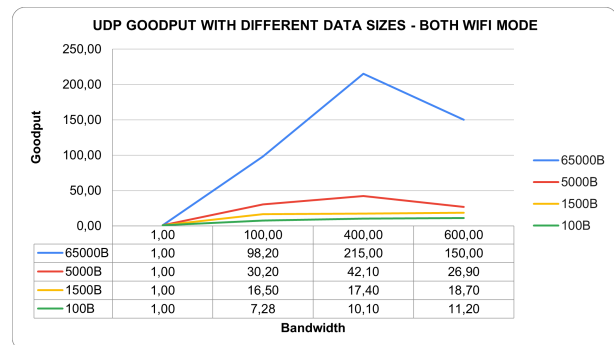


Figure 6: UDP goodput with different application buffer sizes and different bandwidth in "Both Wifi Mode"

The packet loss is also higher, reaching 95% with smaller buffer sizes but being still remarkably high even with a buffer size of 65000B.

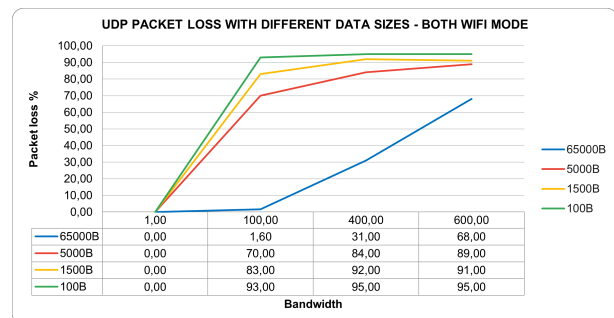


Figure 7: UDP packet loss with different application buffer sizes and different bandwidth in "Both Wifi Mode"

5 CONCLUSION

In conclusion, analysing the two configurations in which two computers are connected via Wi-Fi-WiFi and LAN-WiFi respectively, we

observe that, under ideal lossless conditions, a TCP transmission should theoretically perform worse than UDP. This is because UDP packets have a lower overhead (8 bytes compared to TCP's 20), allowing a theoretically higher throughput. However, in reality, the packet loss and lack of reliability of the UDP protocol make TCP significantly more advantageous, providing a more stable and efficient transmission. Furthermore, analysing the graphs, it can be seen that as the packet size increases, UDP improves its performance: when transmitting over a WiFi network, the collision avoidance mechanism reduces the number of packets transmitted, thus decreasing the risk of collisions and/or packet loss and improving the overall throughput. Finally, it is shown that the LAN-WiFi configuration offers greater efficiency and quality of service than WiFi-WiFi alone. This is because the wired connection in the first part of the link eliminates the problems related to interference and collisions that are typical of completely wireless networks, increasing efficiency as only one round trip time is required, guaranteeing a more stable transmission with less latency.

A APPENDIX AND WRITER'S REMARKS

We would like to thank the reader for the attention given to this analysis and for the interest in comparing different network configurations. During the realisation of this study, one of the main challenges faced was the configuration and integration of Linux and Windows devices, often running on virtual machines, and their connection via LANs, WiFi and, in some cases, hybrid solutions. The adaptation of network settings, driver management and compatibility between the different operating systems required not only a good understanding of the technologies involved, but also a certain amount of **creativity and perseverance** to overcome the inevitable difficulties.

A particularly interesting aspect that emerged during the analysis was the comparison between theoretical performance, derived from ideal calculations, and the results actually obtained in practical tests. As to be expected, the real world introduces a number of limiting factors, including interference, system resource management and unpredictable network protocol behaviour, which led to lower performance than assumed in a perfectly ideal scenario. This underlines the importance of always evaluating theoretical models in the light of concrete experimental tests, in order to obtain a more realistic view of how networks behave under actual operating conditions.